

Cloning Service Now Production instance to Dev

Let's learn how to clone the Service Now Production instance to Dev instance.

Through out this article the **Source Instance is Prod instance** and **Target instance is Dev instance**.

Everything that we are doing below is in Source Instance (PROD) only.

Pre requisites for Cloning:

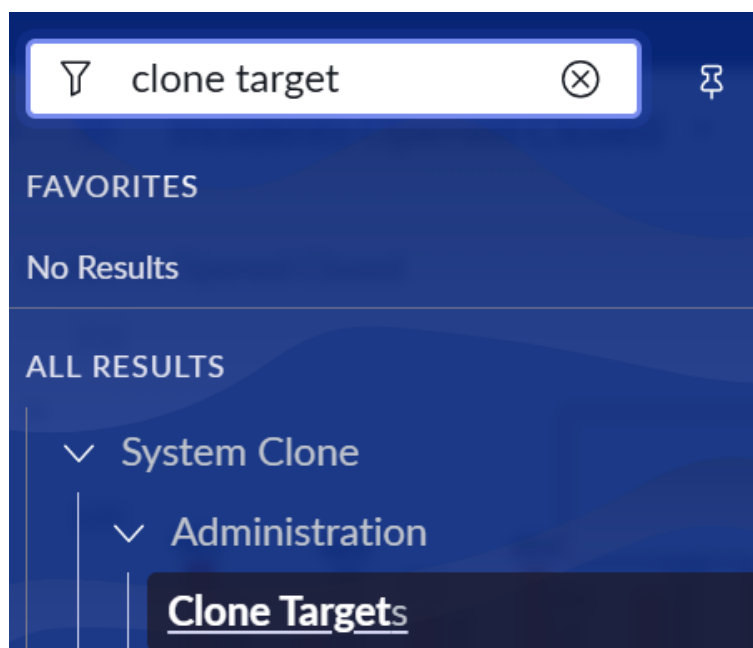
1. **Take the backup from Target (Dev) instance.**
2. **Create a Target Instance Record.**
3. **Create a Clone Profile.**
4. **Create records in Exclude Tables.**
5. **Create records in Preserve Data.**
6. **Schedule a Clone Request.**

Take the backup from Target (Dev) instance.

Always backup all the update sets on the target instance before cloning. Once the cloning is done then you can again load the update sets on the target instance and commit them and start working. This will ensure all the development which is there in the Target instance is safe post cloning.

Create a Target Instance record.

Navigate to **System Clone > Administration > Clone Targets**



Click on **New** to create a new Target Instance record.

ServiceNow Instance
New record View: clone_request

* Instance URL

* Username

* Password

Submit Save

Instance URL: Give the URL of the Target (Dev) instance.

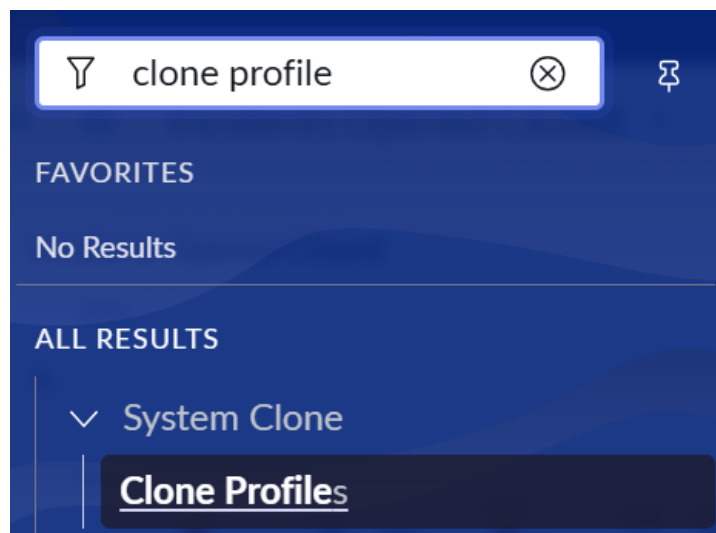
Username: Username of a user from Target (Dev) instance.

Password: Password of the user.

Note: This user in target instance should have the **admin** and **clone_admin** roles.

Create a Clone Profile in Source instance.

Navigate to **System Clone > Clone Profiles**.



Click on **New** to create a clone profile.

Clone Profile
New record View: clone_profile

Preservers, exclusions and cleanup scripts can be added to the profile once it is saved.

* Name Target Instance

Make Default ☐

Options

Exclude Tables Specified In Exclusion List ☒

Exclude Audit And Log Data ☒

Exclude Attachment Data ☒

Preserve Theme ☒

Data Copied From Specific Tables

Preserve In Progress Global Update Sets

Preserve users and related tables ☒

Save Submit

Give it a **Name**, select the **Target Instance** which you created above.

Check all the necessary check boxes below and save the record.

Now It is most important to decide what data to be cloned from Source to Target instance and what data not to be cloned.

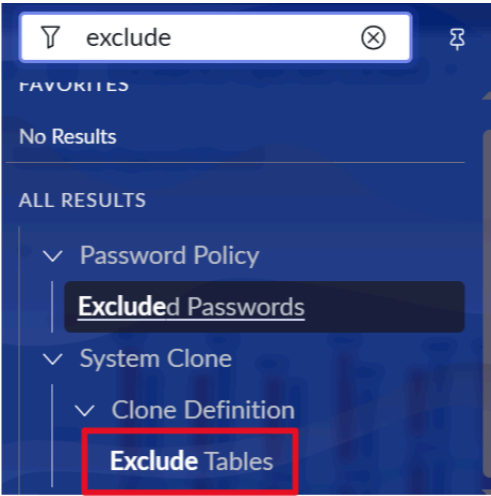
To do this efficiently we need to know the concept of Exclude Tables and Preserve Tables.

Create records in Exclude Tables.

Exclude data is nothing but the data we exclude bringing from Source to the Target instance.

A common use case is to exclude user and groups tables from Source to Target Instance.

To Navigate to exclude data go to **All > System Clone > Clone Definition > Exclude Tables**.



If you want to exclude any table from Source to Target then create a record in this **Exclude Tables**

This will tell the System to bring empty table for whatever tables mentioned in the **Excluded List**.

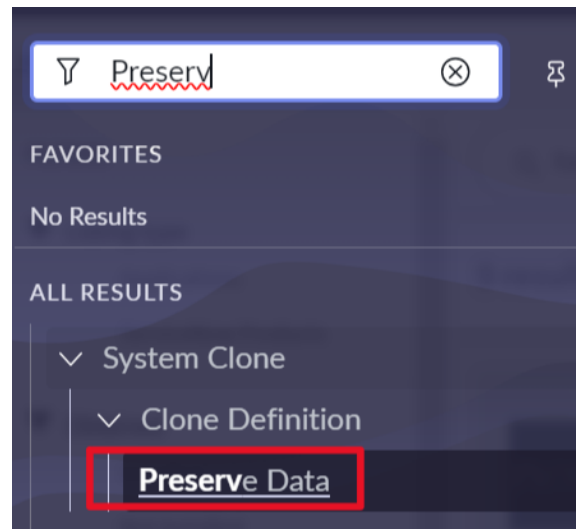
Clone Exclude Tables		
Created	Search	
All > Include in System Clone = true		
Name	Application	Created
Search	Search	Search
sys_alias	Global	13/03/2025 16:16:24
sys_gen_ai_usage_log	Global	06/09/2023 19:12:29
idp_attribute	Global	14/06/2023 06:52:43
sys_cs_channel_user_profile	Global	05/05/2023 08:09:59
sn_instance_clone_request	Clone Admin Console	14/04/2023 21:39:26
sys_user_certificate	Global	13/04/2023 09:35:59
sys_ca_certificate	Global	13/04/2023 09:11:23
ecc_agent_ext_context_vcenter	Global	07/03/2023 08:33:31
ecc_agent_ext_context_trap	Global	07/03/2023 08:30:30
sn_kmf_resolution_framework_log		03/10/2022 20:15:18
sys_analytics_batch_state	Global	28/09/2022 15:56:53
sys_analytics_data_points_error	Global	28/09/2022 08:00:12
sys_analytics_data_points_queue	Global	28/09/2022 07:59:12

These are list of tables that are excluded from cloning the data from Prod to Dev.

Create records in Preserve Data.

Preserve Tables/Preserve Data is nothing but preserving the data on the target instance on selected tables so that it is not overwritten.

To configure this Preserve Data navigate to **All > System Clone > Clone Definition > Preserve Data**.



If you want preserve the data of any table in the target instance, then create a record here mentioning that table.

A screenshot of a web application interface showing a table titled 'Clone Data Preservers'. The table has four columns: 'Name', 'Table', 'Conditions', and 'Theme'. The 'Name' column has a search input field. The 'Table' column has a search input field. The 'Conditions' column has a search input field. The 'Theme' column has a search input field. The table contains several rows of data, including 'Adapter Configuration', 'Adapter Configuration Page', 'Adapter Message', 'AES App Details', 'AES User Project Application Mapping', 'AES User Projects', and 'App Collaboration Descriptor'. The 'Table' column for 'AES User Project Application Mapping' contains the text 'Applications in Projects (Deprecated) [sn_app_eng_studio_project_application_m2m]'. The 'Table' column for 'AES User Projects' contains the text 'Project (Deprecated) [sn_app_eng_studio_project]'. The 'Table' column for 'App Collaboration Descriptor' contains the text 'App Collaboration Descriptor [sn_app_eng_studio_app_collaboration_descriptor]'. The table is paginated, showing '1 to 20 of 111' records.

Name	Table	Conditions	Theme
Adapter Configuration	Configuration [sys_cs_adapter_configuration]		false
Adapter Configuration Page	Configuration Page [sys_cs_adapter_configuration_page]		false
Adapter Message	Adapter Message [sys_cs_adapter_message]		false
AES App Details	App Details [sn_app_eng_studio_app_details]		false
AES User Project Application Mapping	Applications in Projects (Deprecated) [sn_app_eng_studio_project_application_m2m]		false
AES User Projects	Project (Deprecated) [sn_app_eng_studio_project]		false
App Collaboration Descriptor	App Collaboration Descriptor [sn_app_eng_studio_app_collaboration_descriptor]		false

For Example you are trying to preserve data on Target instance for **User Preference** table.

Check if User Preference table is listed in this Preserve Data.

A screenshot of the same 'Clone Data Preservers' table, but with a filter applied. The filter bar at the top shows 'All > Include in System Clone = true > Table contains sys_user_preference'. The table now only shows one row with the 'Table' column containing '*sys_user_preference'. The 'Name' column has a search input field. The 'Conditions' column has a search input field. The 'Theme' column has a search input field.

Name	Table	Conditions	Theme
	*sys_user_preference		

If not create a new record and select User Preference table so that data will be preserved.

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Clone Data Preserver

New record

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Submit

✱ Name

User Preference

Theme

☐

Application

Global

ⓘ

✱ Table

User Preference [sys_user_preference]

▼

Conditions

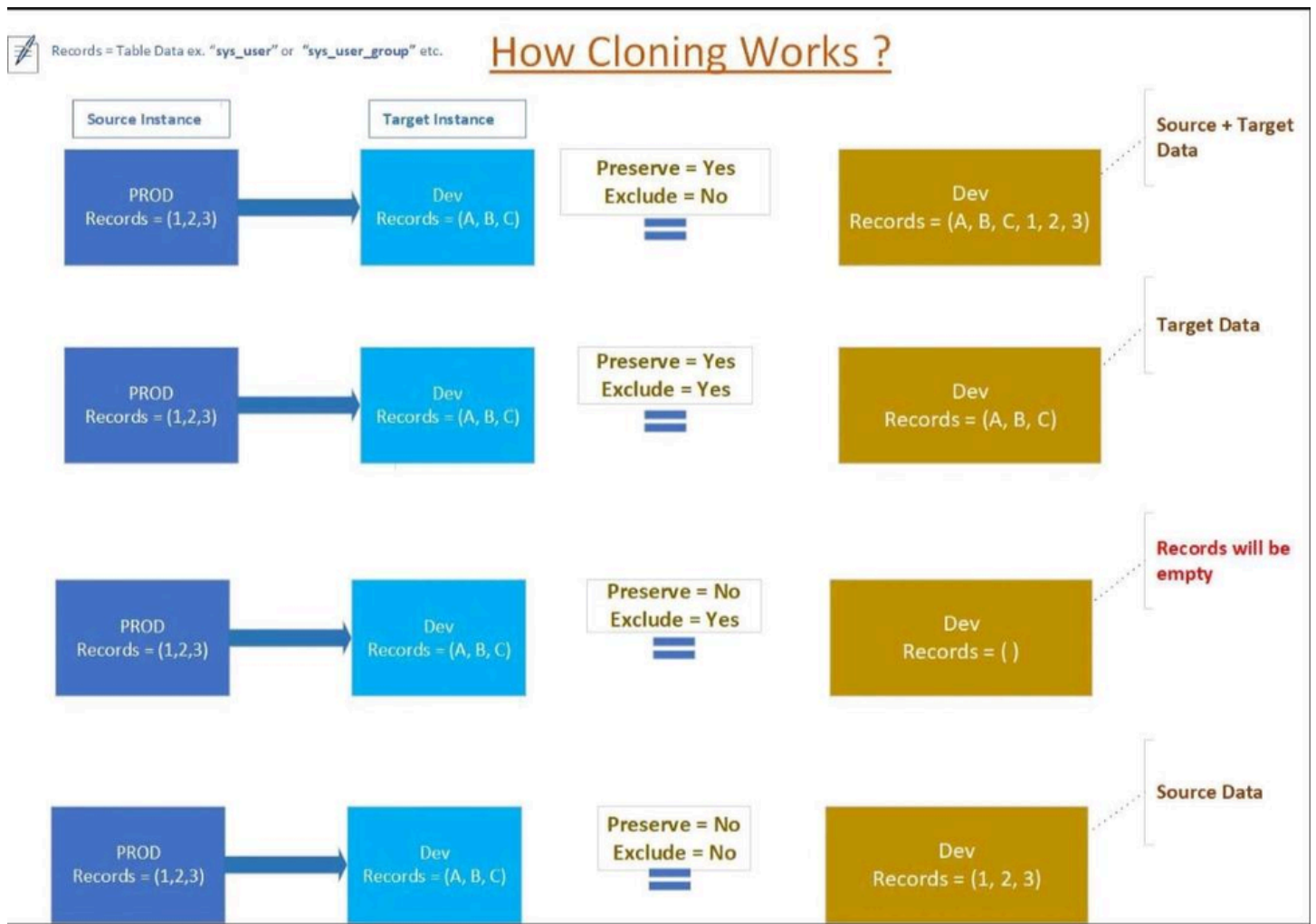
Add Filter Condition

Add "OR" Clause

Submit

We can even add some conditions to define what particular user preferences we need to preserve.

Here is how the Preserve and Exclude table work together.

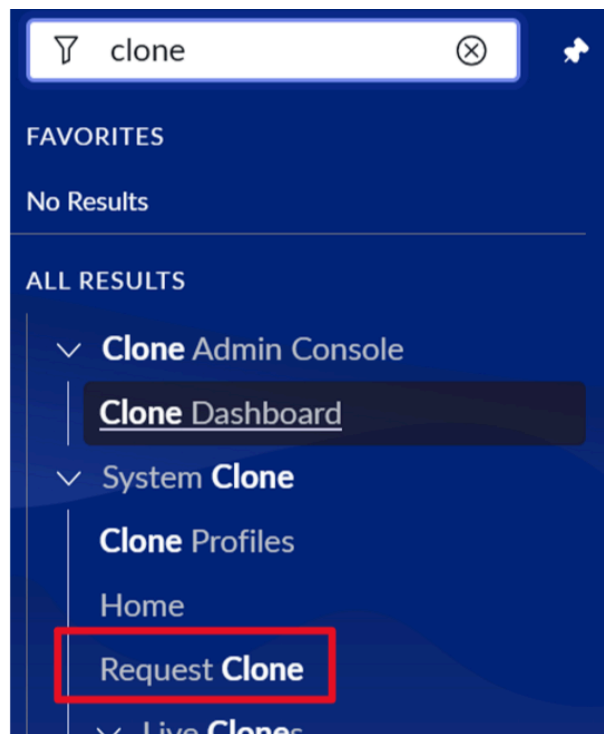


Consider Preserving or Excluding this below tables with Attention.

User Roles, Groups, Group Membership
Group Roles, and Role Contains records

Schedule a Clone Request

Now let's schedule the cloning. To do this, navigate to **System Clone > Request Clone**



Create a new record here.

System Clone
New record View: clone_request

Submit Conflict Calendar Save

① We have re-designed and streamlined the clone experience. Go to the new [\[Clone Admin Console\]](#) to take advantage of the new features.

Cloning copies configuration and application data from the source instance to the target instance ([More info](#)). Backup any unpublished custom applications and update sets before requesting a clone ([More info](#))

Profile

* Target Instance

* Clone Scheduled Start Time

Email upon completion ☐

Options

Submit Conflict Calendar Save

Mention the below details

Profile: Mention the profile which you created above

Target Instance: Mention the target instance which you created above

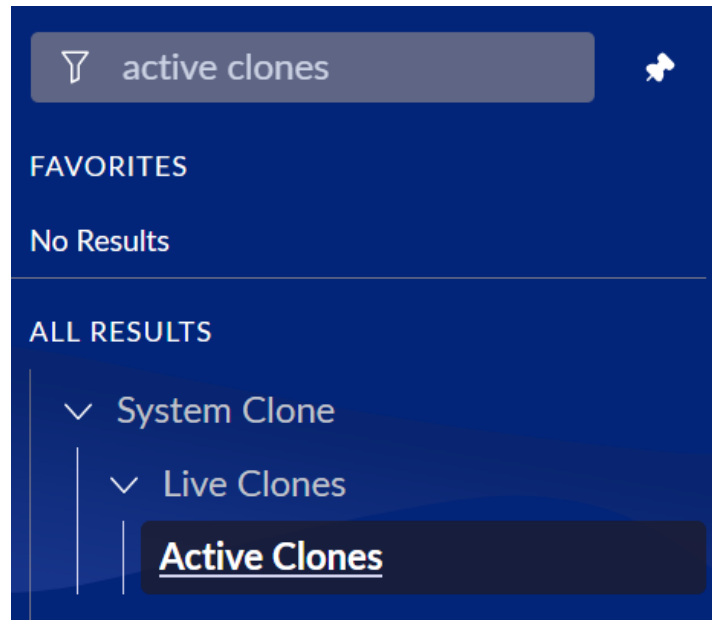
Clone Scheduled Start Time: This will allow us to select the Start time of the schedule. The time selected is evaluated against any conflicts with Target instance like target instance receiving any patch or any Change. In that case you'll be prompted to adjust the start time.

Email Upon Completion: This will allow us to select user who wants to be notified when clone has been completed.

After filling the above mentioned details and click on save, system will Authenticate target instance with username and password you gave above. Once it is authenticated successfully the clone will be scheduled.

You can find the active scheduled clone request here

Go to **System Clone > Live Clone > Active Clone**



Your active clones will be shown here in the list view. Click on that and view the clone logs.

Tips on Cloning:

- If your source instance is on different version from Target instance, then target instance will have the source instance version post cloning.
- Always look for the system property **glide.db.clone.allow_clone_target** is set to **True** in target instance.

glide.db.clone.allow_clone_target = true

- Can you roll back a clone? YES we can do it but its not recommended unless it is unavoidable
 1. Go to System **Clone > Live Clones > Clone History**
 2. Select the clone to be roll back .
 3. Click on roll back under related links and give the reason for roll back.
 4. This must be done within 7 days from clone happened.

